WaterTAP Costing Reference Guide

Detailed Unit Model Costing Parameters

These models compute costs from physics-derived sizing variables. Capital cost equations vary by unit type. All parameters are tunable.

Unit Model	Capital Cost Eq.	Key Default Parameters	Reference
Anaerobic Digester	$A \times (Q/Q_{ref})^B$	A = \$19.36M B = 0.6 Q_{ref} = 911 m^3/day	NREL Waste-to-Energy Model
CSTR	$A \times V^B$	A = \$1,246.1/m^3 B = 0.71	C.C. Tang, 1984
Chiller	$C_{chill} \times (W_{duty}/COP)$	C_{chill} = \$200/kW COP = 7	
Clarifier (Circular)	$A \times A_{surf}^2 + B \times A_{surf} + C$	A = -6×10^{-4} B = 98.952 C = 191,806	Sharma et al. 2013
Clarifier (Primary)	$A \times Q_{MGD}^{B}$	A = \$538,746 B = 0.7	Byun et al. 2022
Clarifier (Rectangular)	$A \times A_{surf}^2 + B \times A_{surf} + C$	A = -2.9×10^{-3} B = 169.19 C = \$94,365	Sharma et al. 2013
Compressor	$C_{comp} \times W_{mech}^{B}$	C_{comp} = \$7364 B = 0.7	El-Sayed et al., 2001
Crystallizer	$A \times (Q/Q_{ref})^B \times IEC$	A = \$675,000 Q_{ref} = 1 m^3/hr B = 0.53 IEC = 1.43 steam = \$0.004/kg	Woods, 2007; Diab and Gerogiorgis, 2017; Yusuf et al., 2019; Panagopoulos, 2019
Dewatering (Centrifuge)	$A \times Q + B$	A = \$328.03/(gal/hr) B = \$751,295	McGivney & Kawamura, 2008
Dewatering (Filter Belt Press)	$A \times Q + B$	A = \$146.29/(gal/hr) B = \$433,972	McGivney & Kawamura, 2008
Dewatering (Filter Plate Press)	$A \times Q^B$	A = \$102,784/(gal/hr) B = 0.4216	McGivney & Kawamura, 2008
Electrocoagulation	$C_{electrodes} + C_{powersupply} + C_{reactor}$	$C_{electrodes}$ = \$2.23/kg (aluminum), \$3.41/kg (iron) $C_{powersupply}$ = linear f(watt) $C_{reactor}$ = power f(vol)	McGivney & Kawamura, 2008; Smith, 2005; Anuf et al., 2022; magna-power.com
Electrodialysis	$C_{mem} \times A_{mem} + C_{electrode}$	C_{mem} = \$160/m^2 $C_{electrode}$ = \$2100/m^2 replacement = 20 %/yr (mem + electrode)	Bian et al., 2018
Electrolyzer	$C_{mem} + C_{anode} + C_{cathode}$	C_{mem} = \$25/m^2 C_{anode} = \$300/m^2 $C_{cathode}$ = \$600/m^2 mat fraction = 65 %	Bommaraju & O'Brien, 2015; Kent (Ed.), 2007; O'Brien, Bommaraju, & Hine, 2005; Yee, 2012
Energy Recovery Device	$C_{ERD} \times Q$	C_{ERD} = \$535/(m^3/s)	
Evaporator	$C_{evap} \times A_{evap}$	C_{evap} = \$1000/m^2 material factor = 1.0	

GAC	$C_{contactor} + C_{gac} + C_{other}$	$C_{contactor}$ = polynomial f(vol) C_{gac} = exponential f(mass) C_{other} = power f(vol) C_{regen} = \$4.28/kg C_{makeup} = \$4.58/kg	EPA-WBS 2021
Heat Exchanger	$C_{HX} \times A_{hx}$	C_{HX} = \$300/m^2 steam = \$0.008/kg	
Heater (Electric)	$C_{heat} \times W_{duty}/eff$	C_{heat} = \$66/kW eff = 0.99	
Ion Exchange (Cation / Anion)	$C_{resin} + C_{vessels} + C_{tanks}$ $C_i = A^b$	$C_{res,AX}$ = \$205/ft^3 $C_{res,CX}$ = \$153/ft^3 vessel A=1596.5, b=0.46 resin replacement = 5 %/yr	EPA-WBS 2021
Membrane Distillation	$C_{mem} \times A_{mem}$	C_{mem} = \$56/m^2 replacement = 20 %/yr	Shamlou, Vidic, & Khanna, 2022
Mixer	$C_{mix} \times Q$	Generic = \$361/(L/s) NaOCl mixer = \$5.08/(m^3/day) $CaOH_2$ = 873.9/(kg/day)	
Nanofiltration	$C_{mem} \times A_{mem}$	C_{mem} = \$15/m^2 replacement = 20 %/yr	
OARO	$C_{mem} \times A_{mem}$	C_{mem} = \$30/m^2 $C_{mem,HP}$ = \$50/m^2 replacement = 15 %/yr	Bartholomew et al. 2017
Pressure Exchanger	$C_{PX} \times Q$	C_{PX} = \$535/(m^3/s)	
Pump (high pressure)	$C_{pump} \times W_{mech}$	C_{pump} = \$53/W	Malek et al. 1996
Pump (low pressure)	$C_{pump} \times Q$	C_{pump} = \$889/(m^3/s)	Bartholomew et al. 2018
Reverse Osmosis	$C_{mem} \times A_{mem}$	C_{mem} = \$30/m^2 $C_{mem,HP}$ = \$75/m^2 replacement = 20 %/yr	Bartholomew et al. 2018
Steam Ejector	$A \times (M)^B$	M = motive steam + entrained vapor; kg/hr A = \$1949 B = 0.3 steam = \$0.008/kg	Gabriel, 2015
Thickener	$A \times A_{surf} + B$	A = 4729.8/ft^2 B = \$37,068	McGivney & Kawamura, 2008
UV+AOP	$C_{reactor} + C_{lamp}$	$C_{reactor}$ = \$202.35/kW C_{lamp} = \$235.50/kW lamp replacement = 33.3 %/yr	UV System Cost Analysis Tool (UVCAT); Wright, Gaithuma, Greene, Aieta, 2006

WaterTAP Costing Reference Guide (continued)

Zero-Order Unit Model Costing — Default Parameters

This table is for zero-order models that use: $C_{cap} = A \times (Q_{in}/Q_{basis})^B$. A is in USD for the reference year shown. Costs are CPI-adjusted to the study year.

Unit Model	A	B	Q_{basis}	Cost Year	Energy (kWh/m³)	Recovery	Reference
Aeration Basin	725,570	0.586	1 MGD	2014	0.412	N/A	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Anaerobic Digestion Oxidation	19,355,231	0.600	911 m³/hr	2012	0.15	0.9999	NREL Waste-to-Energy Model (WESyS)
Backwash Solids Handling	41,812	0.918	1 MGD	2007	f(Q)	0.95	Based on costs for subprocesses in Table 5.7.1 (Equations 29, 30, 31, 32, 33, 37, 38, 39) in Cost Estimating Manual for Water Treatment Facilities (McGivney/Kawamura)
Bio-Active Filtration	725,570	0.586	1 MGD	2014	None	0.99	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Bioreactor	1,252,160	1	45 m³/hr	2018	None	0.97	Developed with Aspen Cost Estimation for San Luis Case Study in WT3.
Cartridge Filtration	725,570	0.586	1 MGD	2014	0.0002	0.9999	Texas Water Development Board IT3P Figure 3.3
Clarifier	177,516	0.318	5,450 m³/hr	2007	None	0.999	McGivney & Kawamura, Figure 5.5.24c + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Conventional Activated Sludge	725,570	0.586	1 MGD	2014	0.14	0.9999	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Cooling Supply	40,299	0.866	1 MGD	2007	None	N/A	McGivney & Kawamura Figure 5.5.35b
Decarbonator	150,000	1	1,577 m³/hr	2007	None	N/A	Survey of High-Recovery and Zero Liquid Discharge Technologies for Water Utilities (2008). Table A2.2.
Fixed Bed Bioreactor	1,542,008	0.734	1 MGD	2017	0.0309	N/A	Regressed from standard designs for EPA-WBS Biological Treatment with Fixed Bed using pressure vessels from 2017
Intrusion Mitigation	44,600,000	1	4,750 m³/hr	1998	0.0512	N/A	Monterey One Case Study Data
Media Filtration	725,570	0.586	1 MGD	2014	0.00015	0.9999	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Membrane Bioreactor	725,570	0.586	1 MGD	2014	f(Q)	0.9999	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Microfiltration	500,000	1	1 MGD	2014	0.18	0.95	Texas Water Development Board IT3PR Table 3.21
Municipal Drinking	40,299	0.866	1 MGD	2007	f(Q)	N/A	McGivney & Kawamura Figure 5.5.35b
Nanofiltration	750,000	1	1 MGD	2014	0.231	0.85	Texas Water Development Board IT3P Table 3.21
Primary Separator	177,516	0.318	5,450 m³/hr	2007	None	0.999	McGivney & Kawamura, Figure 5.5.24c, Cost Estimating Manual for Water Treatment Facilities
Pump	40,299	0.866	1 MGD	2007	0.051	N/A	McGivney & Kawamura Figure 5.5.35b
Screen	7	1	1 m³/d	2018	None	0.9999	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management. Figure 4.6
Seawater Intake	3,624	0.889	1 m³/hr	2018	f(Q)	N/A	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management, Figure 4.2 + Figure 4.4
Settling Pond	177,516	0.318	5,450 m³/hr	2007	None	0.9999	McGivney & Kawamura, Figure 5.5.24c + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities

Static Mixer	1,698	0.325	1 L/s	2010	None	N/A	Chemical Engineering Design, 2nd Edition. Principles, Practice and Economics of Plant and Process Design. Table 7.2 eq fit to power eq.
Surface Discharge	648	0.876	1 m³/d	2020	f(Q)	N/A	https://www.lenntech.es/processes/surface-water-discharge-of-brine.htm
Tri Media Filtration	725,570	0.586	1 MGD	2014	0.00045	0.9	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Ultrafiltration	500,000	1	1 MGD	2014	0.231	0.95	Texas Water Development Board IT3P Table 3.21
Walnut Shell Filter	725,570	0.586	1 MGD	2014	None	0.99	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Water Pumping Station	24,348	0.867	1 MGD	2007	None	N/A	McGivney & Kawamura Figure 5.5.36b

All parameters are defaults and fully tunable by the user. Source: watertap-org/watertap GitHub repository.